

# Abir Harrasse

## PERSONAL INFO

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WEBSITE: [abirharrasse.github.io](https://abirharrasse.github.io)  
GOOGLE SCHOLAR: [scholar.google.com/citations?user=lnaBEG0AAAAJ](https://scholar.google.com/citations?user=lnaBEG0AAAAJ)

## EDUCATION

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| <b>Sep. 2020</b><br>– Sep. 2025 | <b>Master’s Degree in Industrial Management</b><br>EMINES School of Industrial Management, UM6P, Morocco<br>GPA: 3.89/4.0   Minor in Data Science<br><i>Coursework: NLP, LLMs, Machine Learning, Optimization, Statistics.</i><br>Thesis: <i>Tracing Multilingual Representations in LLMs with Cross-Layer Transcoders</i> |
| <b>Sep. 2020</b><br>– Sep. 2024 | <b>Bachelor’s Degree in Industrial Management</b><br>EMINES School of Industrial Management, UM6P, Morocco<br>GPA: 3.96/4.0   Engineering degree program<br><i>Coursework: Real Analysis, Topology, Linear Algebra, Measure Theory, Martingales, Lebesgue Integration, Linear Programming.</i>                             |

## WORK EXPERIENCE

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| <b>Nov. 2025</b><br>– Present   | <b>Research Fellow, Martian Learning Inc.</b> , Bay Area, CA (Remote)<br>Mechanistic interpretability research and uncertainty estimation for LLM routing.                                                                                                    |
| <b>Apr. 2025</b><br>– Sep. 2025 | <b>Research Intern</b> , Max Planck Institute for Intelligent Systems, Tübingen<br>Empirical Inference Lab. Supervised by Prof. Zhijing Jin & Prof. Bernhard Schölkopf.<br>Mechanistic interpretability of multilingual LLMs; training data attribution.      |
| <b>Sep. 2024</b><br>– Mar. 2025 | <b>Research Intern, Martian Learning Inc.</b> , Bay Area, CA (Remote)<br>Developed mechanistic interpretability framework for causal reasoning and cross-model safety intervention transfer.<br>Received return offer for full-time Research Fellow position. |
| <b>Apr. 2024</b><br>– Aug. 2024 | <b>Research Intern</b> , National University of Singapore (Remote)<br>IORA Department.<br>Adversarial multi-agent evaluation systems; 8% performance improvement.                                                                                             |
| <b>Jan. 2023</b><br>– Jan. 2024 | <b>Research Assistant</b> , College of Computing, UM6P, Morocco<br>Supervised by Prof. Omar Saadi.<br>Model-based reinforcement learning; runtime and sampling efficiency.                                                                                    |

## PUBLICATIONS

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### [CLT-Forge: A Scalable Library for Cross-Layer Transcoders and Attribution Graphs](#)

F. Draye, **A. Harrasse**, V. Palit, T.-Y. Wu, J. Liu, P. S. Pandey, R. Wu, T. J. Zhang, Z. Jin, B. Schölkopf.  
*Under Review.*

### [Curveball Steering: The Right Direction To Steer Isn’t Always Linear](#)

S. Raval, H. J. Song, L. Wu, **A. Harrasse**, J. M. Phillips, F. Barez, A. Abdullah.  
*Under Review.*

### [Tracing Multilingual Representations in LLMs with Cross-Layer Transcoders](#)

**A. Harrasse**, F. Draye, P. S. Pandey, Z. Jin, B. Schölkopf.  
*Under Review.*

### [Debate, Deliberate, Decide \(D3\): A Cost-Aware Adversarial Framework for Reliable and Interpretable LLM Evaluation](#)

**A. Harrasse**, C. Bandi, H. Bandi.  
*EACL 2026 (Oral).*

### [STRIDE: Training Data Attribution Can Be Estimated in Activation Space](#)

**A. Harrasse**, R. Dagli, A. Abdullah, Z. Jin.  
*3rd Data-FM Workshop, ICLR 2026.*

**TinySQL: A Progressive Text-to-SQL Dataset for Mechanistic Interpretability Research**

**A. Harrasse**, P. Quirke, C. Neo, D. Nathawani, L. Marks, A. Abdullah.  
*EMNLP 2025.*

**Disentangling and Steering Multilingual Representations: Layer-Wise Analysis and Cross-Lingual Control in LLMs**

**A. Harrasse**, F. Draye, B. Schölkopf, Z. Jin.  
*ICML Actionable Interpretability Workshop, 2025.*

**Activation Space Interventions Can Be Transferred Between Large Language Models**

N. F. Oozeer, D. Nathawani, N. Prakash, M. Lan, **A. Harrasse**, A. Abdullah.  
*ICML 2025.*

## TALKS

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| <b>Aug. 2025</b> | <b>Beyond Black Boxes: Understanding Training Data Influence in Language Models</b><br>Amazon, Tübingen, Germany. |
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## SELECTED AWARDS

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| <b>2026</b>      | ETH AI Center Doctoral Fellowship.                                                         |
| <b>2025</b>      | Best Master's Thesis, EMINES, UM6P.                                                        |
| <b>2025</b>      | First Class Honours, EMINES, UM6P.                                                         |
| <b>2023</b>      | BCG Platinion Hackathon: National Winner, International 2nd Place.                         |
| <b>2020–2025</b> | UM6P Excellence Scholarship: Top 1% of applicants.                                         |
| <b>2020</b>      | National Moroccan Merit Scholarship (FAR): Top 0.1% of candidates.                         |
| <b>2019–2020</b> | NaMO: Selected among the top 25 nationally for Moroccan Mathematical Olympiad preparation. |

## ACADEMIC SERVICE

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| ORGANIZER | Local Chair, AISTATS 2026          |
| REVIEWER  | ICML 2026, COLM 2026, NeurIPS 2026 |

## CIVIC ENGAGEMENT

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|-----------|-----------------------------------------------------|
| 2025      | <b>Founder</b> , <i>National Day of Mathematics</i> |
| 2023–2024 | <b>Board Member</b> , <i>Math&amp;Maroc</i>         |
| 2023      | <b>Founder</b> , <i>ThinkAI Hackathon</i>           |
| 2023–2024 | <b>Organizer</b> , <i>GenAI Winter School</i>       |
| 2022–2024 | <b>Co-founder</b> , <i>Rotaract EMINES</i>          |
| 2021–2023 | <b>Founder</b> , <i>E-Maths Club</i>                |
| 2021      | <b>Participant</b> , <i>Caravane du Sourire</i>     |

## LANGUAGES

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|                           |                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------|
| <b>Natural Languages:</b> | Arabic (Native), French (Native), English (Fluent), Turkish (Intermediate), Mandarin (HSK 1) |
| <b>Programming:</b>       | Python, PyTorch, Transformers, SLURM                                                         |

*Last updated April 2026*